

A counterexample to the blocked-class bound as stated

Concerning G. Duran-Ballester's Erdős–Gyárfás working paper, Zenodo v1

Scope. This note disproves the quantitative estimate on the entire class explicitly defined in `def:blocked-class`. It is not a counterexample to the Erdős–Gyárfás conjecture. Node [171] assumes [170]; the point is a mismatch between the defined counting class and the scope required by that estimate, not a refutation of a conditional implication under arbitrary additional hypotheses.

1. The estimate under examination. The defined class $\mathcal{B}(\mathcal{P})$ consists of labelled n -vertex, m -edge graphs with minimum degree at least three, containing the fixed induced P_{13} windows, such that no power-of-two cycle passes through any window. Node [171], `lem:blocked-graphs-compress`, proposes, conditional on [170],

$$|\mathcal{B}(\mathcal{P})| \leq |\mathcal{G}_{n,m}| n^{-(c_{13}-o(1))p_{13}}, \quad c_{13} = 118.108581006\dots \quad (1)$$

Here $|\mathcal{G}_{n,m}| \leq \binom{n}{m}$. The locations used are the author-supplied TeX labels `def:blocked-class`, `lem:scale-additivity` [170], and `lem:blocked-graphs-compress` [171]. Source: DOI 10.5281/zenodo.22019344, v1, published 20 August 2026; file `erdos_64_proof.tex`.

2. A five-vertex attachment. Let J be K_4 with one edge subdivided once. Its vertices are s, u, v, w, z and its edges are

$$su, sv, uw, uz, vw, vz, wz.$$

The root s has degree two; all other vertices have degree three. Attaching s by one edge to an external vertex makes every vertex of J cubic. An induced path in J starting at s has at most three vertices: after s, u, w , either remaining choice creates a chord (sv or uz); the other cases are symmetric.

3. A connected cubic block on 66 vertices. Take seven disjoint triangles $T_i = \{a_i, b_i, c_i\}$, $1 \leq i \leq 7$, and add the six edges $b_i a_{i+1}$, $1 \leq i \leq 6$. At each of the nine degree-two vertices

$$a_1, b_7, c_1, \dots, c_7$$

attach a separate copy of J by one edge to its root. Call the resulting graph H . It is finite, simple, connected and cubic, with

$$n(H) = 7 \cdot 3 + 9 \cdot 5 = 66, \quad m(H) = 99.$$

Choose the window

$$P = a_1, b_1, a_2, b_2, \dots, a_6, b_6, a_7.$$

These 13 vertices induce exactly a path. The six inter-triangle edges and all nine gadget attachments are bridges. A simple cycle cannot use a bridge, so every cycle meeting P is contained in one triangle and has length three. Thus P is blocked at every power-of-two scale, even under the strong interpretation that a cycle “passes through” a window whenever it meets any window vertex.

4. The maximum induced- P_{13} packing number of H is one. An induced path uses at most two vertices from each triangle. A path meeting a pendant gadget and the rest of H visits that gadget only at an endpoint, using at most three gadget vertices; at most two such gadgets can be visited.

Delete the central triangle T_4 . Each side component has three triangles, and an induced path in it has at most

$$3 \cdot 2 + 2 \cdot 3 = 12$$

vertices. The other component is a five-vertex gadget. Hence every induced P_{13} meets T_4 . If it used only one vertex of T_4 , that vertex would be an endpoint: each vertex of T_4 has just one neighbour outside T_4 . The path would then have at most

$$1 + 3 \cdot 2 + 3 = 10$$

vertices (and fewer on the central gadget side). Consequently every induced P_{13} uses at least two of the three vertices of T_4 . Any two such paths intersect. Since the displayed P exists, the maximum packing size is exactly one. This proves more than inclusion-maximality.

5. An unbounded family contradicting (1). Let H_p be the disjoint union of p copies of H , with one displayed window per copy. Its window family $\mathcal{P}_p$ is maximum, and

$$n = 66p, \quad m = 99p = \frac{3}{2}n, \quad p_{13} = p.$$

Fix labels and these window positions. Then $H_p \in \mathcal{B}(\mathcal{P}_p)$ by the stated definition, so $|\mathcal{B}(\mathcal{P}_p)| \geq 1$. No connectedness requirement is present in that definition. The family is exactly cubic, so the numerical near-cubic condition is satisfied with zero degree surplus.

The elementary inequality $\binom{N}{m} \leq (eN/m)^m$ gives

$$\binom{\binom{n}{2}}{99p} \leq \left(\frac{e(n-1)}{3} \right)^{99p} < n^{99p},$$

since $e < 3$. Substitution into (1) would therefore give

$$1 \leq |\mathcal{B}(\mathcal{P}_p)| < n^{(99-118.108581006\dots+o(1))p} < 1$$

for every sufficiently large p , a contradiction. The fixed negative margin exceeds 19 in the coefficient of $p \log n$; a term tending to zero cannot remove it. No numerical cutoff or extrapolation from a finite search is used.

6. What follows, and what does not. The estimate cannot hold for the entire class specified by `def:blocked-class`. Nodes [170] and [171] cannot together establish it on that whole class. The graphs have 4-cycles inside the pendant gadgets, away from the windows, and are not minimal counterexamples to the conjecture. They are not claimed to satisfy every earlier residual or no-exit condition. Restricting the class to such survivors could exclude them, but would change the counting domain and require a correspondingly scoped proof.

Question for the author. Which condition excludes H_p from the class actually counted in [171]? If a narrower survivor class is intended, where is that class defined and the compression estimate established for it?

Verification. Two separately implemented constructions agree. Exhaustive checks find 208 unoriented induced P_{13} paths, each meeting T_4 in exactly two vertices. These checks support the finite block construction; the all- p argument is the proof above and requires no code. This note was prepared with AI assistance.